TASK 2 — DATA RETRIEVAL

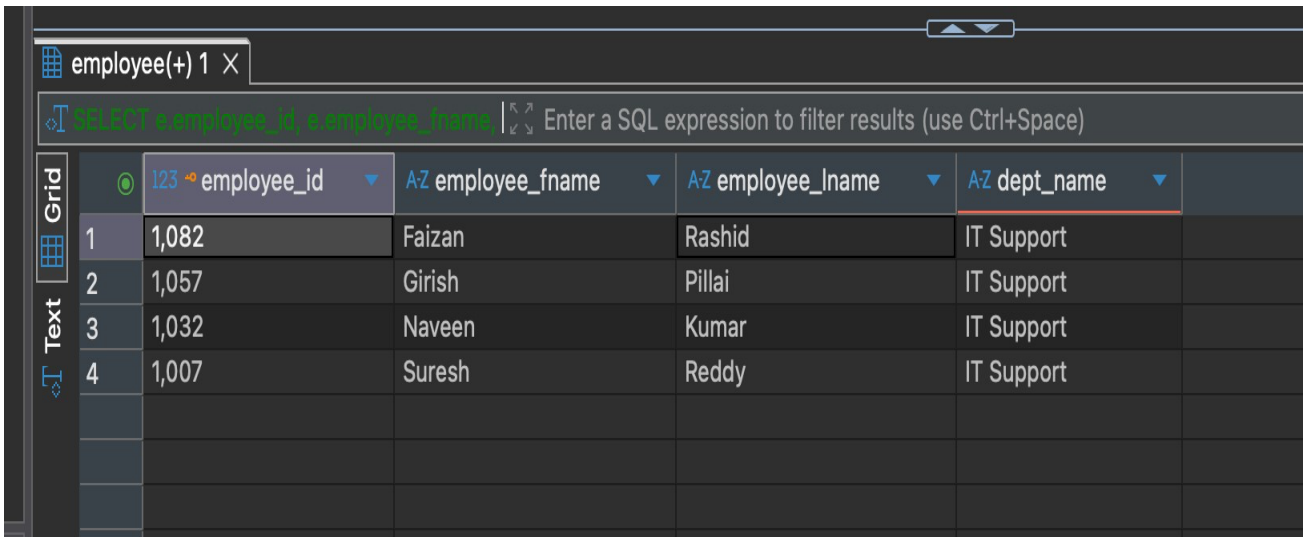
Aim for the task: Retrieve specific records based on conditions.

Q1) List all employees working in IT-related departments.

Query:

```
SELECT e.employee_id,
       e.employee_fname,
       e.employee_lname,
       d.dept_name
FROM employee e
JOIN department d
ON e.dept_id = d.dept_id
WHERE d.dept_name LIKE '%IT%';
```

Explanation:
This query retrieves the employee ID, first name, and last name from the employee table along with the department name from the department table. It uses an inner join on the dept_id column and filters results to include only employees whose department name contains the keyword “IT”.



	employee_id	employee_fname	employee_lname	dept_name
1	1,082	Faizan	Rashid	IT Support
2	1,057	Girish	Pillai	IT Support
3	1,032	Naveen	Kumar	IT Support
4	1,007	Suresh	Reddy	IT Support

Q2) List all employees working on any project.

Query:

```
SELECT e.employee_id,
       e.employee_fname,
       e.employee_lname,
```

```
    t.team_id,
    p.project_name
FROM employee e
JOIN team t ON e.team_id = t.team_id
JOIN project p ON t.project_id = p.project_id;
```

Explanation:
This query displays employees who are assigned to projects. It joins the employee table with the team table using team_id and then joins the project table using project_id. The output includes employee details, team ID, and the associated project name.

employee(+) 1

SELECT e.employee_id, e.employee_fname, Enter a SQL expression to filter results (use Ctrl+Space)

	employee_id	employee_fname	employee_lname	team_id	project_name
1	1,001	Amit	Sharma	401	Website Redesign
2	1,002	Riya	Patel	402	Mobile Application
3	1,003	Rahul	Verma	403	ERP Deployment
4	1,004	Neha	Iyer	404	Cloud Migration
5	1,005	Karan	Mehta	405	Data Warehouse
6	1,006	Priya	Nair	406	Cybersecurity Audit
7	1,007	Suresh	Reddy	407	CRM Upgrade
8	1,008	Ananya	Sen	408	AI Chatbot
9	1,009	Vikram	Singh	409	E-commerce Portal
10	1,010	Pooja	Gupta	410	Payment Integration
11	1,011	Arjun	Kapoor	411	HR Automation
12	1,012	Mehul	Choksi	412	Inventory System
13	1,013	Tanvi	Kulkarni	413	Analytics Dashboard
14	1,014	Rakesh	Mishra	414	Networking Upgrade
15	1,015	Nitin	Aggarwal	415	IoT Monitoring
16	1,016	Sonal	Desai	416	Digital Marketing
17	1,017	Mohit	Malhotra	417	Payroll System
18	1,018	Ayesha	Khan	418	Document Management

Q3) List all projects started in 2024 along with client names.

Query:

```
SELECT p.project_id,
    p.project_name,
    p.start_date,
    c.client_fname,
    c.client_lname
FROM project p
JOIN client c
ON p.client_id = c.client_id
WHERE p.start_date >= '2024-01-01';
```

Explanation:
This query retrieves projects that started in the year 2024 or later. It joins the project table with the client table using client_id and displays the project details along with the client's first and last names.

project(+) 1

SELECT p.project_id, p.project_name, p.sta | Enter a SQL expression to filter results (use Ctrl+Space)

	project_id	project_name	start_date	client_fname	client_lname
1	301	Website Redesign	2024-01-05	Amit	Sharma
2	302	Mobile Application	2024-02-01	Riya	Patel
3	303	ERP Deployment	2024-01-20	Rahul	Verma
4	304	Cloud Migration	2024-03-10	Neha	Iyer
5	305	Data Warehouse	2024-02-15	Karan	Mehta
6	306	Cybersecurity Audit	2024-04-01	Priya	Nair
7	307	CRM Upgrade	2024-03-05	Suresh	Reddy
8	308	AI Chatbot	2024-05-01	Ananya	Sen
9	309	E-commerce Portal	2024-01-25	Vikram	Singh
10	310	Payment Integration	2024-02-10	Pooja	Gupta
11	311	HR Automation	2024-03-15	Arjun	Kapoor
12	312	Inventory System	2024-01-30	Mehul	Choksi
13	313	Analytics Dashboard	2024-02-20	Tanvi	Kulkarni
14	314	Networking Upgrade	2024-04-10	Rakesh	Mishra
15	315	IoT Monitoring	2024-05-05	Nitin	Aggarwal
16	316	Digital Marketing	2024-01-10	Sonal	Desai
17	317	Payroll System	2024-02-25	Mohit	Malhotra
18	318	Document Management	2024-03-01	Ayesha	Khan

Q4) Find all employees belonging to the ‘Software Development’ department.

Query:

```
SELECT e.employee_id,
       e.employee_fname,
       e.employee_lname,
       d.dept_name
FROM employee e
JOIN department d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'Software Development';
```

Explanation:
This query lists employees who work specifically in the Software Development department. It joins the employee and department tables on dept_id and filters the results to return only employees whose department name matches “Software Development”.

employee(+) 1

SELECT e.employee_id, e.employee_fname, | Enter a SQL expression to filter results (use Ctrl+Space)

	employee_id	employee_fname	employee_lname	dept_name
1	1,008	Ananya	Sen	Software Development
2	1,033	Ishita	Ghosh	Software Development
3	1,058	Hina	Parveen	Software Development
4	1,083	Gautam	Roy	Software Development